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## Architectures, Standards and Engineering Library (A.R.S.EN.L)

### ARSENL Contents

1. [ARSENL Home \(/programs/arsenl/index.php\)](/programs/arsenl/index.php)
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4. [Reference Architectures \(/programs/arsenl/gra.php\)](/programs/arsenl/gra.php)
5. [Contact Us \(mailto:HQAFCEN.DigitalCampaign@us.af.mil\)](mailto:HQAFCEN.DigitalCampaign@us.af.mil)

The A.R.S.EN.L provides a central body of knowledge and references for architecture. This repository provides insight into Government Reference Architectures, Modular Open Systems Approach (MOSA) Enabling Standards, and information about Air Force System Interfaces.

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## Architectures, Standards and Engineering Library (A.R.S.EN.L) ARSENL Access Instructions

ARSENL Contents

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### How to get access to the full Architectures and Standards Engineering Library (A.R.S.EN.L) Content including CUI information

AFMC has established an A.R.S.EN.L page on the Virtual Distributed Laboratory (VDL) website to support dissemination of Air Force Government Reference Architectures, Modular Open Systems Approach Enabling Standards, and Interface information. An account on VDL will provide you with access to all Distribution Limited and Controlled Unclassified information.

To get access, please follow the instructions below:

#### VDL Instructions:

1. Please go to the public VDL page at <https://www.vdl.af.mil/> (<https://www.vdl.af.mil/>) to request an account. The system will request some basic contact information and a sponsor. Government and non-Government personnel requesting access must provide a Government sponsor (e.g., Government Program Manager). The sponsor will be contacted via email to verify that the requester is qualified to receive Distribution Limited and International Traffic in Arms Regulations (ITAR) material. The sponsor will need to confirm requester's status via email or phone before access to the site is granted.

Virtual Distributed Laboratory


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2. Once you have VDL access, please go to the restricted VDL page at <https://restricted.vdl.af.mil/> (<https://restricted.vdl.af.mil/>) to request access to the ARSENL VDL project. Click "Programs" and "Request Program Access". Scroll down to find the "ARSENL" project and select "Request Access". The system will automatically notify approval authorities and a Government ARSENL representative will approve your access.

#### Points of Contact:

Sponsor – Air Force Material Command (AFMC):

[HQAFMCEN.DigitalCampaign@us.af.mil](mailto:HQAFMCEN.DigitalCampaign@us.af.mil) (<mailto:HQAFMCEN.DigitalCampaign@us.af.mil>)

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# Architectures, Standards and Engineering Library (A.R.S.EN.L) Standards

ARSENL Contents

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The Following List of Publicly Releasable MOSA Enabling Standards Support USAF MOSA Strategy

- [Technical Standard for Future Airborne Capability Environment \(FACE\) \(/programs/arsenl/files/FACE Technical Standard v3.1.pdf\)](/programs/arsenl/files/FACE_Technical_Standard_v3.1.pdf)
  - This document embodies a set of requirements and descriptions referred to as the FACE Technical Standard. The FACE Technical Standard uses industry standards for distributed communications, programming languages, graphics, operating systems, and other areas as appropriate. The FACE Technical Standard begins with an architectural overview introducing the Reference Architecture, followed by a detailed description of each architectural segment and interface. This document then outlines requirements of each software component of the Reference Architecture. Finally, the appendices list the specific Application Programming Interfaces (APIs) required by the FACE Technical Standard, the FACE Data Architecture required by the FACE Technical Standard, as well as other applicable standards. Specific FACE terms are defined in the Glossary.
- [Sensor Open Systems Architecture \(SOSA\) \(/programs/arsenl/files/SOSA Technical Standard Edition 2.0.pdf\)](/programs/arsenl/files/SOSA_Technical_Standard_Edition_2.0.pdf)
  - The goal of The Open Group SOSA Consortium is to develop open architecture at the right level for Communications (Comms), Electro-Optical/Infra-Red (EO/IR), Electronic Warfare (EW), Radar, and Signals Intelligence (SIGINT) systems. The open architecture supports airborne, subsurface, surface, ground, and space. The SOSA Consortium strives to develop an ecosystem that allows interoperability, reuse, and faster delivery of products to market through vertical integration from cables, mechanical interfaces, hardware, software, and system designs.
- [Software Communication Architecture \(/programs/arsenl/files/SCA\\_4.1\\_SCASPECIFICATION.pdf\)](/programs/arsenl/files/SCA_4.1_SCASPECIFICATION.pdf)
  - The Software Communications Architecture (SCA) is published by the Joint Tactical Networking Center (JTNC). This architecture was developed to assist in the development of Software Defined Radio (SDR) communication systems, capturing the benefits of recent technology advances which are expected to greatly enhance interoperability of communication systems and reduce development and deployment costs.
- [Red Hawk \(/programs/arsenl/files/REDHAWK-2.3 SCA Specification.pdf\)](/programs/arsenl/files/REDHAWK-2.3_SCA_Specification.pdf)
  - REDHAWK is a software-defined radio (SDR) framework designed to support the development, deployment, and management of real-time software radio applications. To support the design and

development of software applications, REDHAWK provides tools that allow development and testing of software modules called "Components" and composition of Components into "Waveform Applications" that can be seamlessly deployed on a single computer or multiple network-enabled computer.

- [Common Payload Interface Standard \(CoPaIS\) \(/programs/arsenl/files/CoPaIS - IEEE-ISTO\\_MOSA-IF-S-001.pdf\)](/programs/arsenl/files/CoPaIS - IEEE-ISTO_MOSA-IF-S-001.pdf)
  - The Common Payload Interface Standard (CoPaIS) standardizes the space vehicle bus-to-payload interface. While SMC-SIS-001 support both ground-based and on-orbit bus-to-payload integration, the other portions (e.g., mechanical, thermal, etc.) are specifically for on-orbit bus-to-payload integration. The CoPaIS payload can be small to large mission payloads on the nadir side of the bus or a modular propulsion subsystem on the zenith side of the bus.

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